



EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



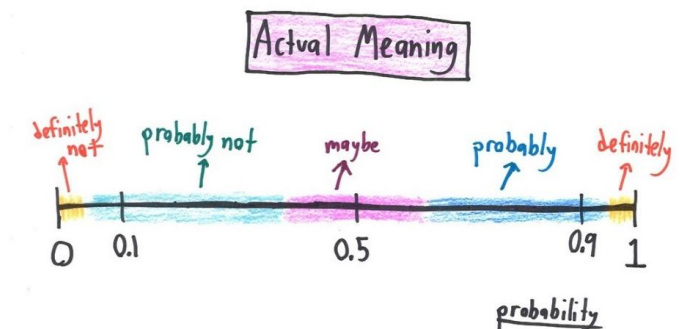


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

aamir hasan

Week 16 - Lecture No 29 – December 01, 2025





Announcements!

1. All lectures posted.
2. Assignment No 3 – by December 05, 2025
3. Quiz No 10 – December 03, 2025
4. End-term exam – Thursday, December 11, 2025 (@Tariq Rafi Hall)

Course Code	Section	Course Title	Date of Examination	Time Start	Time End	Room No.	Enrollment	Name of Instructor(s)
EE/CE 354/361	L1 L2 L3	Introduction to Probability and Statistics	Thursday, December 11, 2025	8:30a	10:30a	E-220	72	Aamir Hasan Tabsheer Ali Askari

Agenda for today

1. Unit 5 – Continuous Random Variables

➡ ➤ Bayes Rule ($X \rightarrow Y$)

1. Discrete $\rightarrow$ Discrete
2. Discrete $\rightarrow$ Continuous
3. Continuous $\rightarrow$ Discrete
4. Continuous $\rightarrow$ Continuous

The Bayes Rule

Discrete Setting

$X \longrightarrow Y$

unobserved value x

observed value y

Prior $P_X(\cdot)$

Model $P_{Y|X}(\cdot|\cdot)$

Think of a Digital Communication Signal



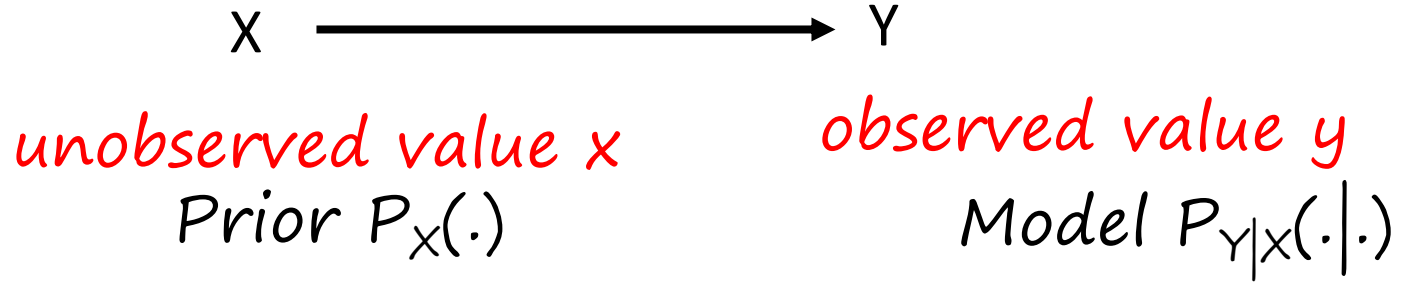
x
unknown state

*noisy observation
of y*

Inference $P_{X|Y}(\cdot|y)$

The Bayes Rule

another example of application

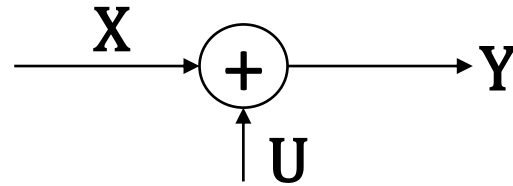


Medical tests – say COVID – 19 test



The Bayes Rule

Example ($Y = X + U$)



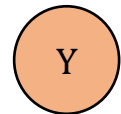
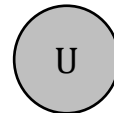
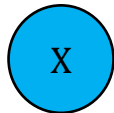
X

Y

unobserved value x
Prior $P_X(\cdot)$

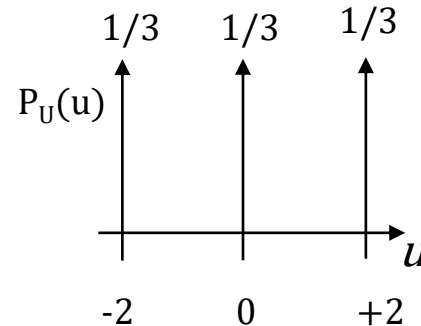
observed value y
Model $P_{Y|X}(\cdot|\cdot)$

Inference $P_{X|Y}(\cdot|y)$



-1 $p_1 = 0.5$

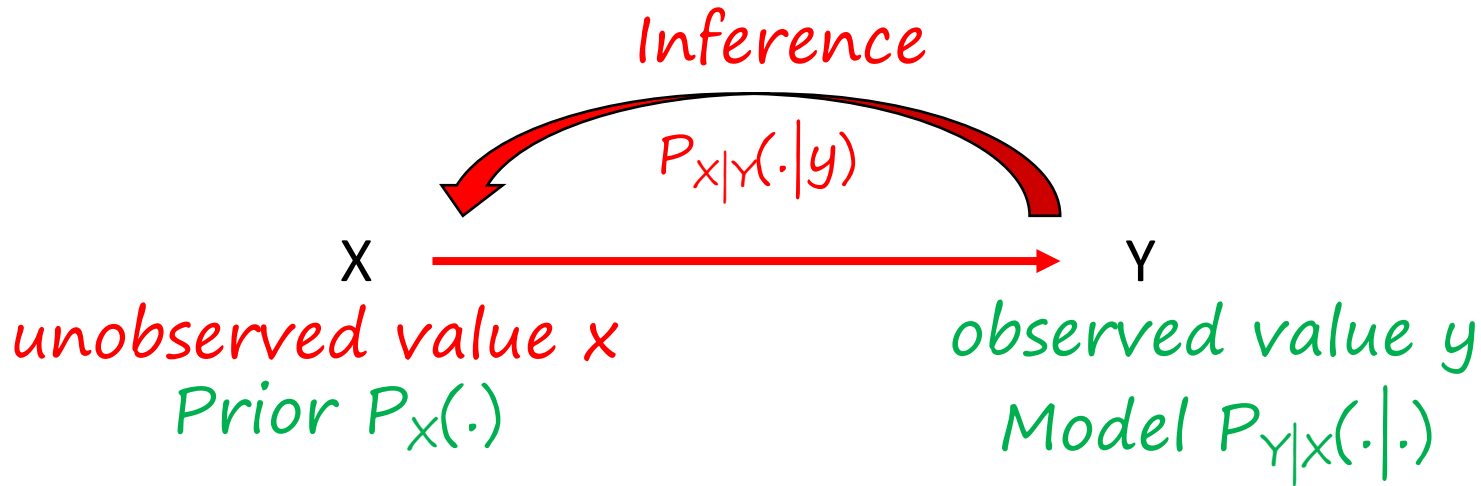
+1 $p_2 = 0.5$



When $X = -1$
 Y can be -3, -1 & +1

When $X = +1$
 Y can be -1, +1 & +3

The Bayes Rule – a theme with variations



$$p_{X,Y}(x, y) = p_Y(y) p_{X|Y}(x|y)$$

$$p_{X,Y}(x, y) = p_X(x) p_{Y|X}(y|x)$$

$$p_{X|Y}(x|y) = \frac{p_X(x) p_{Y|X}(y|x)}{p_Y(y)}$$

$$p_Y(y) = \sum_x p_X(x) p_{Y|X}(y|x)$$

$$f_{X,Y}(x, y) = f_Y(y) f_{X|Y}(x|y)$$

$$f_{X,Y}(x, y) = f_X(x) f_{Y|X}(y|x)$$

$$f_{X|Y}(x|y) = \frac{f_X(x) f_{Y|X}(y|x)}{f_Y(y)}$$

$$f_Y(y) = \int f_X(x) f_{Y|X}(y|x) dx$$

The Bayes Rule – one discrete and one continuous RV

- K: discrete
- Y: continuous

$$P(K=k, y \leq Y \leq y+\delta) = \delta > 0, \delta \approx 0$$

$$P(K=k) \cdot P(y \leq Y \leq y+\delta \mid K=k) \approx P_K(k) f_{Y|K}(y|k) \delta$$

$$P(K=k \mid y \leq Y \leq y+\delta) \cdot P(y \leq Y \leq y+\delta) \approx f_Y(y) \delta P_{K|Y}(k|y)$$

$$P_{K|Y}(k|y) = \frac{P_K(k) f_{Y|K}(y|k)}{f_Y(y)}$$

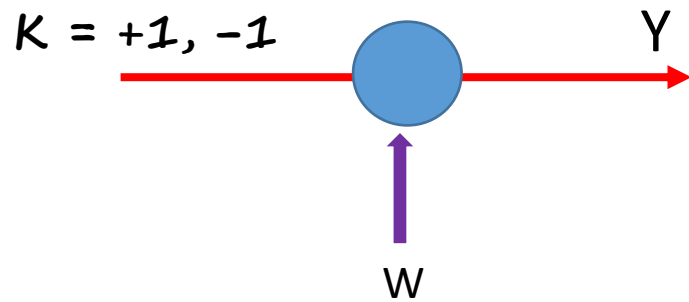
$$f_{Y|K}(y|k) = \frac{f_Y(y) P_{K|Y}(k|y)}{P_K(k)}$$

$$f_Y(y) = \sum_{k'} p_K(k') f_{Y|K}(y|k')$$

$$p_K(k) = \int f_Y(y') p_{K|Y}(k|y') dy'$$

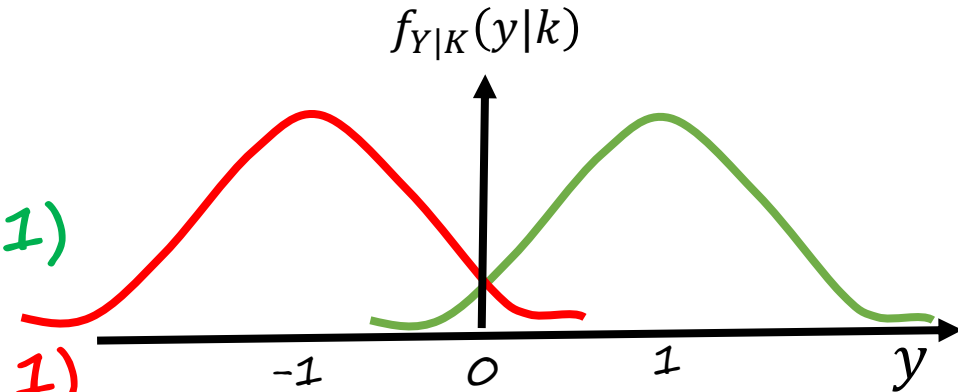
The Bayes Rule – discrete unknown and continuous measurement

- Unknown K : equally likely to be $+1, -1$
- measurement Y : $Y = K + W$ $W \sim N(0, 1)$



$$Y | K = +1 \sim N(1, 1)$$

$$Y | K = -1 \sim N(-1, 1)$$



- Probability that $K = +1$, given that $Y = y$? $P_{K|Y}(1|y)$

- $P_K(k) = 1 / 2$

$$K = +1, -1$$

- $f_{Y|K}(y|k) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y-k)^2}$

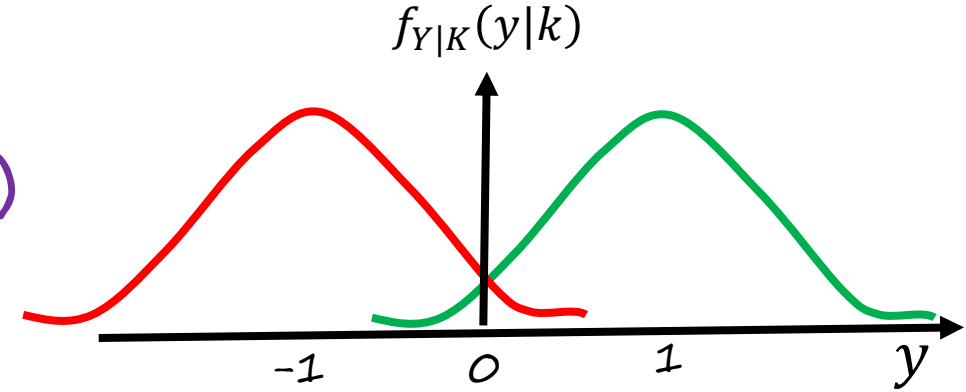
$$P_{K|Y}(k|y) = \frac{P_K(k) f_{Y|K}(y|k)}{f_Y(y)}$$

$$f_Y(y) = \sum_{k'} p_K(k') f_{Y|K}(y|k')$$

The Bayes Rule – discrete unknown and continuous measurement

- Unknown K : equally likely to be $+1, -1$
- measurement Y : $Y = K + W$
- Probability that $K = 1$, given that $Y = y$?

$$W \sim N(0, 1)$$



$$P_{K|Y}(k|y) = \frac{P_K(k) f_{Y|K}(y|k)}{f_Y(y)}$$

$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}(y+1)^2} + \frac{1}{2\sqrt{2\pi}} e^{-\frac{1}{2}(y-1)^2}$$

$$f_Y(y) = \sum_{k'} p_K(k') f_{Y|K}(y|k')$$

$$P_{K|Y}(+1|y) = \frac{1}{1 + e^{-2y}}$$

Interpretation???

